ITA Classroom Activity 4

Date: 25.07.2025 (Friday)

Hour: 1 Hour

Will You Carry the Umbrella?

Consider a situation where we observe the **weather conditions** and whether **people carry an umbrella**. Based on observational data collected over several days, we derive the following **joint probability distribution** of these two discrete random variables:

- **Weather (W)**: {Rainy, Sunny}
- **Umbrella (U)**: {Yes, No}

	Umbrella = Yes	Umbrella = No
Rainy	0.5	0.1
Sunny	0.2	0.2

1. What is the marginal probability that the weather is Rainy?

2. What is the marginal probability that a person carries an umbrella?

3. What is the most likely pair (Weather, Umbrella) to occur?

4. What does the joint entropy represent in this context?

5. Which of the following represents the correct formula for Joint Entropy H(W, U)?

   a) $-\sum_{w} P(w)\log_2 P(w)$
   b) $-\sum_{u} P(u)\log_2 P(u)$
   c) $-\sum_{w,u} P(w,u)\log_2 P(w,u)$
   d) $P(w)+P(u)$

6. What is the joint entropy H(W, U)?

7. Which pair contributes the most to the joint entropy?

   a) Rainy & Yes
   b) Sunny & No
   c) Sunny & Yes
   d) Rainy & No

8. If someone is not carrying an umbrella, the most likely weather is?

9. Compute the following:

   I. H(W)
   II. H(U)
   III. Verify: $H(W,U) \leq H(W)+H(U)$

Self-Learning:

10. If the umbrella decision was completely independent of the weather, would the joint entropy be **higher**, **lower**, or the **same**? Explain and prove your reasoning.

****

The joint distribution table (given):

Weather \ Umbrella	Yes	No
Rainy	0.5	0.1
Sunny	0.2	0.2

Check: total $= 0.5 + 0.1 + 0.2 + 0.2 = 1.0$.

1) Marginal probability that the weather is Rainy

$$P(\text{Rainy}) = P(\text{Rainy,Yes}) + P(\text{Rainy,No}) = 0.5 + 0.1 = 0.6.$$

Answer: $P(\text{Rainy}) = 0.6$ (60%).

2) Marginal probability that a person carries an umbrella

$$P(U = \text{Yes}) = P(\text{Rainy,Yes}) + P(\text{Sunny,Yes}) = 0.5 + 0.2 = 0.7.$$

Answer: $P(\text{Umbrella} = \text{Yes}) = 0.7$ (70%).

3) Most likely pair (W, U)

Look at joint probabilities: 0.5 (Rainy,Yes), 0.1 (Rainy,No), 0.2, 0.2. The largest is **0.5**.

Answer: **(Rainy, Yes)** — probability 0.5.

4) What does the joint entropy represent here?

Joint entropy $H(W, U)$ = the expected information (average surprise) when we observe the pair (Weather, Umbrella).

Concretely: it is the average value of $-\log_2 P(W = w, U = u)$ over the joint distribution. Units: bits.

Interpretation in this context: *on average*, how many bits are needed to encode (or how uncertain we are about) the combined outcome (weather + umbrella decision) before observing it.

5) Correct formula for joint entropy

Options:

a) $-\sum_w P(w) \log_2 P(w)$ — this is $H(W)$.

b) $-\sum_u P(u) \log_2 P(u)$ — this is $H(U)$.

c) $-\sum_{w,u} P(w, u) \log_2 P(w, u)$ — this is the **joint** entropy.

d) $P(w) + P(u)$ — nonsense for entropy.

Answer: (c) $H(W, U) = -\sum_{w,u} P(w, u) \log_2 P(w, u)$.

6) Compute the joint entropy $H(W, U)$

We compute cell contributions $-p \log_2 p = p \cdot (-\log_2 p)$ for each joint cell, then sum.

Known log values we'll use:

- $\log_2(0.5) = -1$.
- $\log_2(0.2) = \log_2(1/5) = -\log_2(5) \approx -2.321928094887362$.
- $\log_2(0.1) = \log_2(1/10) = -\log_2(10) \approx -3.321928094887362$.

Now compute contributions (digit-by-digit):

Now compute contributions (digit-by-digit):

1. $p = 0.5$ (Rainy,Yes):
   $-0.5 \log_2 0.5 = -0.5(-1) = 0.5$.
2. $p = 0.1$ (Rainy,No):
   $-0.1 \log_2 0.1 = -0.1(-3.321928094887362) = 0.3321928094887362$.
3. $p = 0.2$ (Sunny,Yes):
   $-0.2 \log_2 0.2 = -0.2(-2.321928094887362) = 0.4643856189774724$.
4. $p = 0.2$ (Sunny,No): same as previous = 0.4643856189774724.

Sum them:

$$H(W, U) = 0.5 + 0.3321928094887362 + 0.4643856189774724 + 0.4643856189774724.$$

Add stepwise:

- $0.5 + 0.3321928094887362 = 0.8321928094887362$.
- Add first $0.4643856189774724 \to 1.2965784284662086$.
- Add second $0.4643856189774724 \to 1.7609640474443681$.

Answer: $H(W, U) \approx$ **1.7609640474** bits (≈ **1.761 bits**).

7) Which pair contributes the most to the joint entropy?

Compare the four contributions we computed:

- (Rainy, Yes): **0.500000**
- (Rainy, No): 0.332193
- (Sunny, Yes): 0.464386
- (Sunny, No): 0.464386

The largest single contribution is **0.5** from **(Rainy, Yes)**

Answer: **(a) Rainy & Yes** contributes the most to joint entropy.

(Intuition: although (Rainy,Yes) is the least surprising per-event compared with very small probability events, its high probability 0.5 multiplies the per-event surprise and results in the largest total contribution $p(-\log p)$.)

8) If someone is not carrying an umbrella (U = No), which weather is most likely?

We want $P(\text{Weather} \mid U = \text{No})$. First compute $P(U = \text{No})$:

$$P(U = \text{No}) = 0.1 + 0.2 = 0.3.$$

Then conditional probabilities:

- $P(\text{Rainy} \mid U = \text{No}) = \dfrac{P(\text{Rainy,No})}{P(U = \text{No})} = \dfrac{0.1}{0.3} = \dfrac{1}{3} \approx 0.3333$.
- $P(\text{Sunny} \mid U = \text{No}) = \dfrac{0.2}{0.3} = \dfrac{2}{3} \approx 0.6667$.

Answer: Sunny is most likely (probability $2/3$) if someone is not carrying an umbrella.

9) Compute: I. $H(W)$, II. $H(U)$, III. Verify $H(W, U) \le H(W) + H(U)$

I. $H(W)$

Marginals: $P(\text{Rainy}) = 0.6,\ P(\text{Sunny}) = 0.4$.

$$H(W) = -[0.6 \log_2 0.6 + 0.4 \log_2 0.4].$$

Compute logs (identities):

- $\log_2 0.6 = \log_2(3/5) = \log_2 3 - \log_2 5 \approx 1.584962500721156 - 2.321928094887362 =$
- $\log_2 0.6 = \log_2(3/5) = \log_2 3 - \log_2 5 \approx 1.584962500721156 - 2.321928094887362 = -0.736965594166206$.
- $\log_2 0.4 = \log_2(2/5) = -1.321928094887362$.

Now compute:

- $-0.6 \log_2 0.6 = -0.6(-0.736965594166206) = 0.4421793564997236$.
- $-0.4 \log_2 0.4 = -0.4(-1.321928094887362) = 0.5287712379549449$.

Sum:

$$H(W) = 0.4421793564997236 + 0.5287712379549449 = 0.9709505944546685 \text{ bits.}$$

II. $H(U)$

Marginals: $P(U = \text{Yes}) = 0.7,\ P(U = \text{No}) = 0.3$.

$$H(U) = -[0.7 \log_2 0.7 + 0.3 \log_2 0.3].$$

Compute logs:

- $\log_2 0.7 \approx -0.5145731728297583$.
- $\log_2 0.3 = \log_2(3/10) = \log_2 3 - \log_2 10 \approx 1.584962500721156 - 3.321928094887362 = -1.736965594166206$.

Now compute:

- $-0.7 \log_2 0.7 = -0.7(-0.5145731728297583) = 0.3602012209808308$.
- $-0.3 \log_2 0.3 = -0.3(-1.736965594166206) = 0.5210896782498619$.

Sum:

$$H(U) = 0.3602012209808308 + 0.5210896782498619 = 0.8812908992306927 \text{ bits.}$$

III. Verify $H(W, U) \le H(W) + H(U)$

We found:

- $H(W, U) \approx 1.7609640474443681$ bits.
- $H(W) + H(U) \approx 0.9709505944546685 + 0.8812908992306927 = 1.8522414936853612$ bits.

Compare:

$$H(W, U) = 1.7609640 \ \le\ 1.8522415 = H(W) + H(U).$$

Difference (mutual information):

$$H(W) + H(U) - H(W, U) \approx 0.0912774462416807 \text{ bits.}$$

This difference is the **mutual information** $I(W; U)$, which is nonnegative and here positive (~0.0913 bits) — so W and U are slightly dependent.

Answer: inequality holds (numerically verified). $I(W; U) \approx 0.09128$ bits.

10) Self-learning: if umbrella decision were *completely independent* of weather, would the joint entropy be higher, lower, or the same?

Short answer: It would be **higher** (or the **same** only if they already are independent). More precisely, independence implies $H(W, U) = H(W) + H(U)$. In general $H(W, U) = H(W) + H(U) - I(W; U)$, and mutual information $I(W; U) \ge 0$, with equality iff independent. So if dependence is removed ($I(W; U) = 0$), the joint entropy increases to the sum of the marginals.

Proof sketch:

- By definition $I(W; U) = D_{\text{KL}}(P_{W,U} \,||\, P_W P_U) \ge 0$.
- Also $I(W; U) = H(W) + H(U) - H(W, U)$. Rearranging:

$$H(W, U) = H(W) + H(U) - I(W; U).$$

- If W and U are independent, $I(W; U) = 0$ and thus $H(W, U) = H(W) + H(U)$.
- If they are currently dependent with positive I, removing dependence increases $H(W, U)$ by exactly that mutual information amount.

Apply to our data: currently $I(W; U) \approx 0.09128$ bits, so the independent joint entropy would be

$$H(W) + H(U) \approx 1.8522414937 \text{ bits,}$$

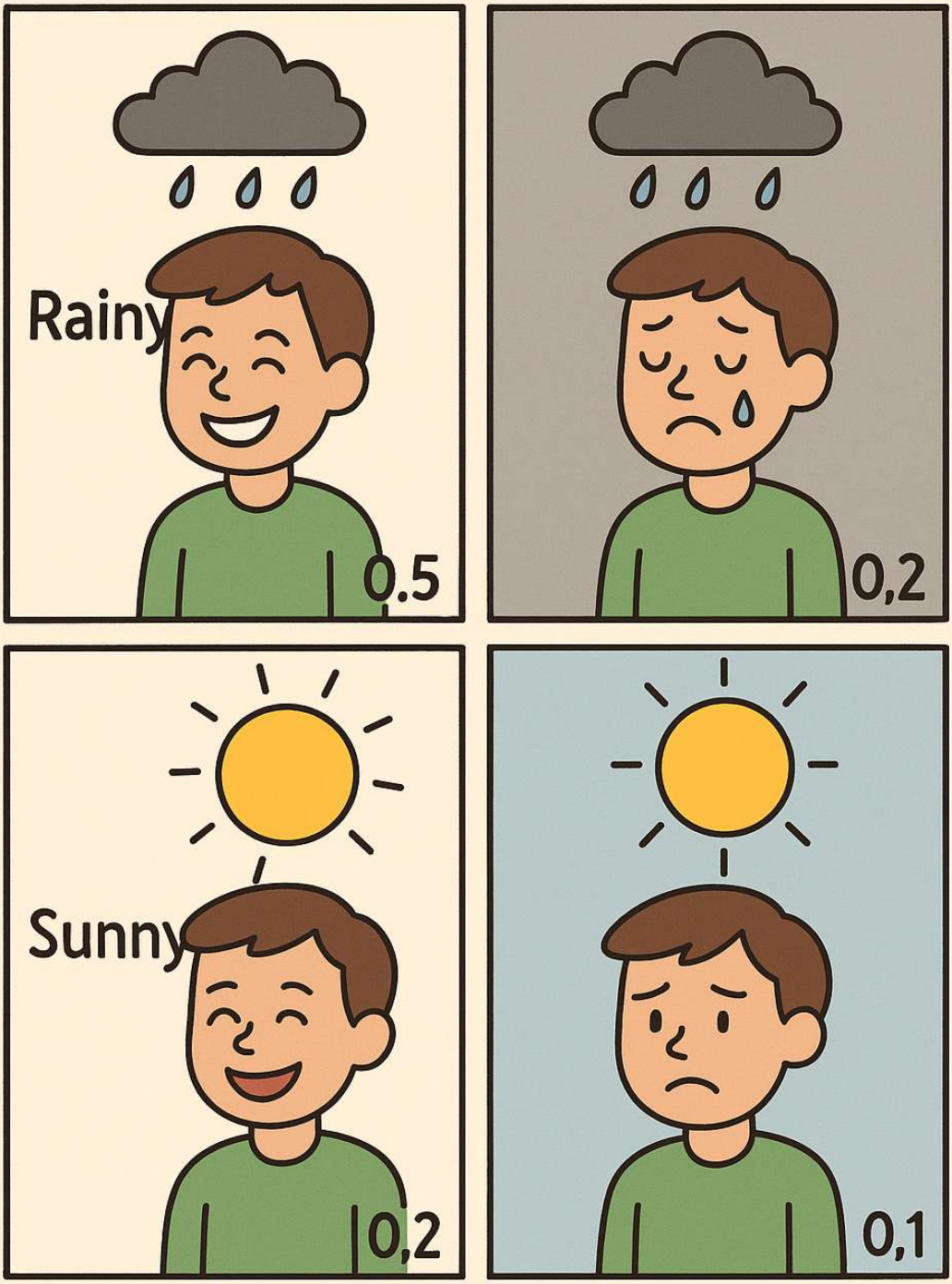
which is about 0.09128 bits larger than the actual joint entropy 1.7609640 bits. So *independence would increase the joint entropy* in this case.

Intuition: dependence means knowing one variable reduces uncertainty about the other. When they are independent you have no such reduction, so the joint uncertainty (bits needed to describe both) is larger.

Extra useful conditional numbers (quick summary)

- $P(U = \text{Yes} \mid \text{Rainy}) = 0.5/0.6 = 5/6 \approx 0.8333$. (Umbrella very likely when rainy.)
- $P(U = \text{Yes} \mid \text{Sunny}) = 0.2/0.4 = 1/2 = 0.5$.
- Conditional entropies:
  - $H(W \mid U) = H(W, U) - H(U) \approx 1.7609640474 - 0.8812908992 = 0.8796731482$ bits.
  - $H(U \mid W) = H(W, U) - H(W) \approx 1.7609640474 - 0.9709505945 = 0.7900134529$ bits.
- Mutual information:

$$I(W; U) = H(W) + H(U) - H(W, U) \approx 0.0912774462 \text{ bits.}$$

This is the average reduction in uncertainty about W when you observe U (and vice versa).

WEATHER VS. MOOD